Anàlisi Global d'Esperança de vida.

1. Problem Statement

In recent decades, the world has undergone an unprecedented demographic and epidemiological transition. As nations develop economically and their populations grow, a double burden emerges: on the one hand, there is greater control over health, but on the other hand, new risks arise, such as the increase in obesity.

In this project, we address different situations that we want to show in a clear and visual way, trying to break down life expectancy into different parameters such as infant mortality or the wealth of the country.

We want to identify which of these factors (health expenditure, fertility, economy) are the best predictors of longevity and how we can classify countries according to their health profiles to support public policy decision-making. We will not carry out a detailed analysis country by country, but we will provide progression maps by country, and from there each country can use the information as it sees fit.

2. Dataset Overview

To carry out this study, we use 3 unified databases from official international sources, covering the period from 1985 to 2016.

Data Sources:

- **Health Nutrition and Population Statistics (World Bank):** Data on life expectancy, child mortality (<5 years), fertility rate, health expenditure (% of GDP and per capita), and Gross National Income (GNI).
- **Obesity among adults (OMS):** Data on the prevalence of obesity (BMI ≥ 30) standardized by age and sex.
- **World Population**: Data on the world's population over the last 50 years.

Processing and Exploratory Data Analysis (EDA):

Data Transformation: We performed data transformation to obtain a single combined dataset and to format the columns in the way that best suits our analysis.

- **Merging:** Unification of the 3 sources using year and country as the common keys.
- **Handling missing values:** Identification of countries with incomplete time series and filtering for specific visualizations (such as the growth chart).

- **Correlations:** The initial correlation matrix revealed a strong positive relationship between Wealth and Life Expectancy (+0.57) and a strong negative relationship between Fertility and Life Expectancy (−0.86). A worrying positive correlation (+0.60) was detected between Life Expectancy and Obesity.

3. Business Questions and Objectives

The main objective is to provide an analytical tool that allows users (researchers or policymakers) to understand the global health situation.

Preguntes Clau:

1. **Health vs. Wealth:** Is GDP a key element for life expectancy?
2. **The Obesity Paradox:** Is it inevitable that obesity increases as a country improves its life expectancy? What causes more deaths: hunger or excess access to food?
3. **Temporal Trends:** Which countries have made the most progress over the last 30 years? Where is the world heading?

4. Methodology

Modelling and Advanced Analysis (Python & Scikit-Learn):

In the notebook EDA_ML.ipynb, we applied Machine Learning techniques to predict life expectancy from country-level indicators. We trained regression models (Random Forest, with a Linear Regression baseline) using socio-economic, demographic and health-related features such as under-5 mortality, fertility rate, health expenditure, income level (GNI per capita), population and obesity prevalence. We evaluated the models using both a random train–test split and a time-based split (training on years ≤ 2005 and testing on > 2005), and analysed feature importances to understand which variables contribute most to explaining life expectancy.

Interactive Visualization (Tableau):

First, we focused on exploring how the world has evolved. We created several dynamic plots that change over time. We believe these are very important because they allow countries to explore where they are heading. They also help identify which improvements should be prioritized, since we have broken down life expectancy by different factors.

Then we created plots to observe how the different predictors behave with respect to the target, and how they evolve. They also help to show that the prediction is not made by an algorithm “just because”; the correlation exists and can be seen very clearly.

The Tableau plots are interactive: you can select years, pause, and change the animation speed for map visualizations. The predictor-vs-target comparison plots allow users to inspect specific data points by clicking on them and to either visualize all countries or select specific ones.

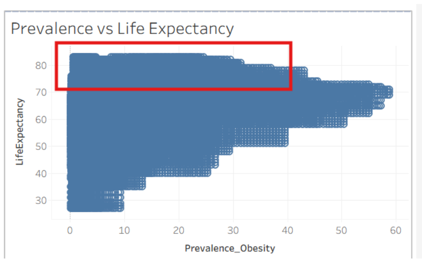
Results Presentation (Streamlit): We developed a web application in Streamlit to democratise access to the model. The app allows users to interactively set key parameters (e.g. under-5 mortality, health expenditure, fertility rate, income level, population, obesity prevalence) and obtain real-time predictions of life expectancy. The interface also integrates the main findings from the EDA (trends and country comparisons) together with the trained model, making the results easier to explore and interpret.

5. Conclusions

The Development Trap:

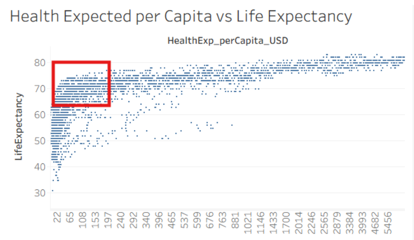
Our analysis confirms a positive association between life expectancy and adult obesity prevalence: countries with higher life expectancy tend to have higher obesity rates on average. This pattern arises clearly in the scatterplots relating life expectancy and obesity, especially among middle- and high-income countries.

The feature importance analysis from the Random Forest model shows that obesity plays a secondary role compared to other drivers: under-5 mortality, income (GNI per capita) and health expenditure explain a much larger share of the variance in life expectancy. In other words, obesity is part of the development picture, but it is not yet the dominant global factor, but we also see in the plot in red that the countries with higher life expectancy don't have high indices in obesity which confirms that for developed countries is an important factor. If there is a day that the globalization arrives equally to all countries then it will be an important factor.



Efficiency beyond Wealth:

The analysis shows that wealth is a very important factor, but the resulting function is very interesting because it is not linear; it looks more like a logarithmic function. As we can see, the countries with the highest economic levels and the greatest investment in health have the highest life expectancy, but there are a few countries which, with limited resources, are still able to achieve very high life expectancy. In our notebook, this appears clearly in the scatter plot of life expectancy versus log_GNI_perCapita_Atlas_USD, where the curve flattens at high income levels, and in the Random Forest feature importances (without under-5 mortality), where GNI and health expenditure emerge as key predictors.



Fertility and Under-5 Mortality Factor:

The reduction in the fertility rate and child mortality are the factors most strongly linked to the increase in life expectancy. Child mortality, to begin with, significantly lowers the average life expectancy, so it is quite a logical factor. Fertility is not as directly related, but a very clear relationship can still be observed. In these cases, the plots are more linear, which is why we added the trend line. In the global temporal maps we can see the evolution and therefore it is easy to predict where we are heading: towards a generally higher life expectancy. In our notebook this is reflected in the very strong negative correlation and almost linear scatter plot between under-5 mortality and life expectancy, as well as in the Random Forest feature importance (where under-5 mortality dominates) and in the alternative model without this variable, where fertility becomes one of the main predictors.

Final Thoughts

We believe that we have done a good study in terms of global trends and human evolution. It would be very interesting now to look separately at economically developed and undeveloped countries to observe how the importance of features changes. We believe that if anything has limited us it is the lack of data in old years and the accuracy of these since measuring some of the metrics in old years was complicated.